Final Project Demo

Vocalytics:
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Recap of Aims and Objective

Project Goal

- Test whether smartphone breathing audio can estimate Respiratory Rate (RR) and VO_2 max
- Compare audio results to smartwatch VO_2 max and Rockport 10 Mile walk test (backup method)

Objective

- RR Accuracy: Detect RR from audio during a 2-min walk at 3 speeds (target ±3–5 BPM).
- VO_2 max Correlation: Compare audio features to VO_2 max from smartwatch or Rockport test (target $r \geq 0.40$).
- Feasibility: Determine if microphones can estimate aerobic fitness within ~10–15% of VO_2 max.

Background & Research Gap

What we know

- Cardiopulmonary Exercise Test (CPET) = gold standard (Hill et al., 1924)
- Wearable vary in accuracy : 10-15% error (Henriksen 2020; Lambe 2025; Jamieson 2024)
- Algorithms are opaque + inconsistent (Shei et al., 2022)

Our Research

- Use everyday devices to test RR and VO_2max trends
- Simple in-ear Bluetooth mic + phone + smartphone

Methods Changes to the plan

Devices Used

- Smartphone + phyphox (audio amplitude)
- Smartwatch + Smart Ring

Protocols

- 3 x 2 minutes walks: easy, medium, hard
- 20 minutes walk/ run for smartwatch VO2 max update

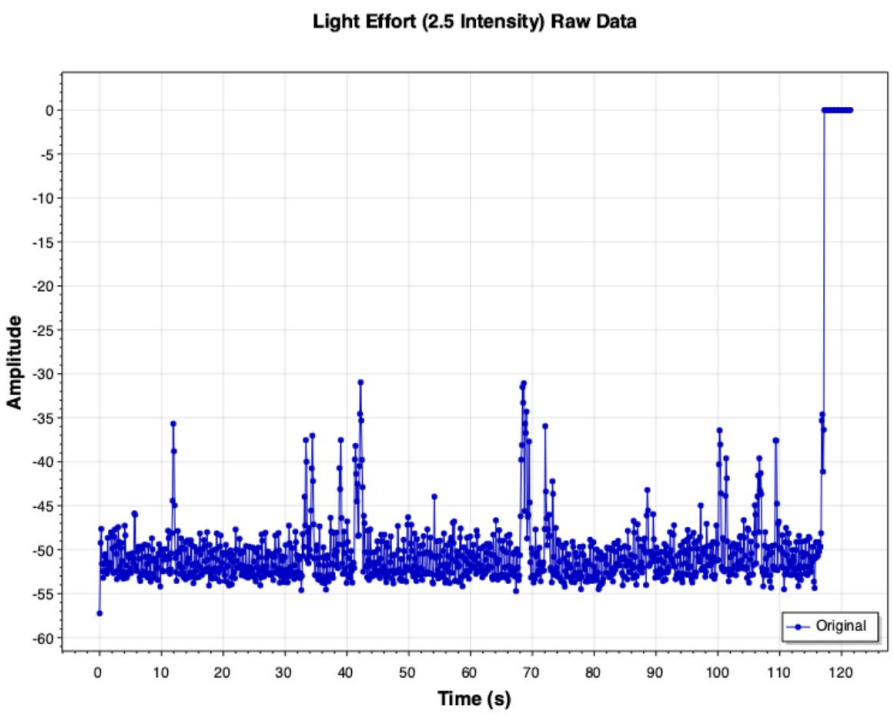
Changes from Original Plan

- Dropped long 20-minute audio recording(Unrealistic)

Raw Data Plot

Amplitudes

Time (s)	Sound pressure level (dB)
0E+00	-5.724355697E+01
1.58436125E-01	-4.922215642E+01
2.629709583E-01	-4.763983498E+01
3.673544583E-01	-5.160323589E+01
4.69524375E-01	-5.246100603E+01
5.759737083E-01	-5.320749932E+01
6.816632916E-01	-5.163787331E+01
7.864614167E-01	-5.072213554E+01
8.922785E-01	-5.050146163E+01
9.985617917E-01	-5.128269707E+01
1.103198542E+00	-5.230551132E+01
1.204294208E+00	-5.271133647E+01
1.30989E+00	-4.865260991E+01
1.415419333E+00	-5.163578815E+01



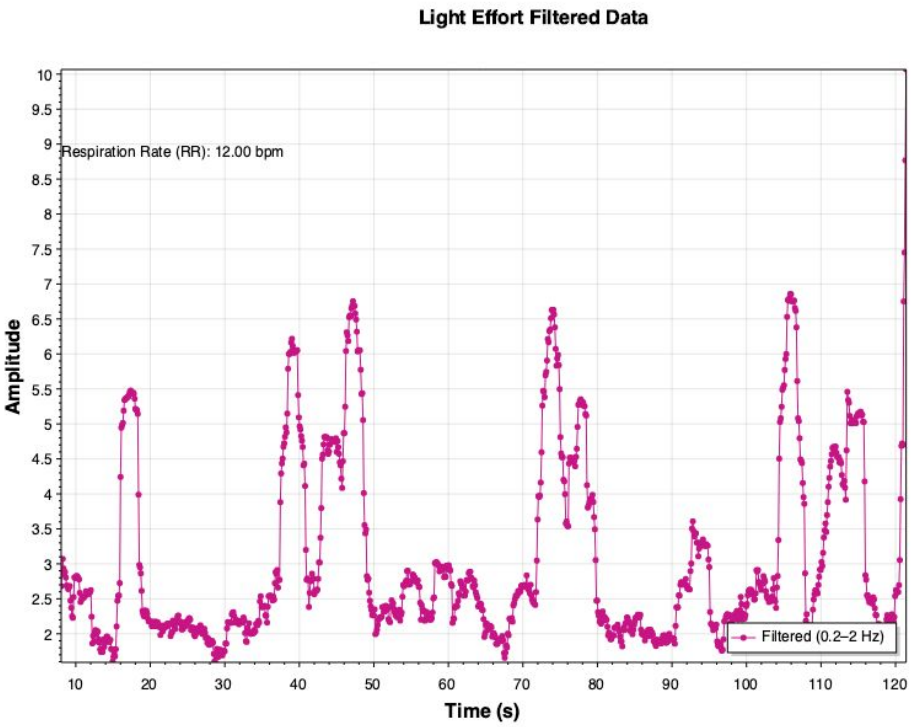
Filtered Data Plot

Languages/Packages Utilized

- .NET Core - C#
- CsvHelper
- NWaves
- ScottPlot packages

What's Going on in the Code

- We calculated the sampling rate
- Applied a band-pass filter
- Smoothed the data using moving RMS
- Determined light, moderate, and heavy breathing levels
- Counted the RR by counting the peaks and converting peak count to breaths/min
- Added a start and end time filter



Data Filtering Methods

Band-Pass Filter:

- This method removes low-frequency drift and high frequency noise using a Finite Impulse Response (FIR) to remove frequencies outside of the standard breathing range, 0.2-2Hz (Lyons, et al., 2011)

Moving Root Mean Square (RMS):

- This method applies sliding root mean square window over the length of the signal, smoothing and normalizing sharp peaks allowing for easier peak detection to determine respiration rate (Smith, 1999).

Calculating Respiration Rate:

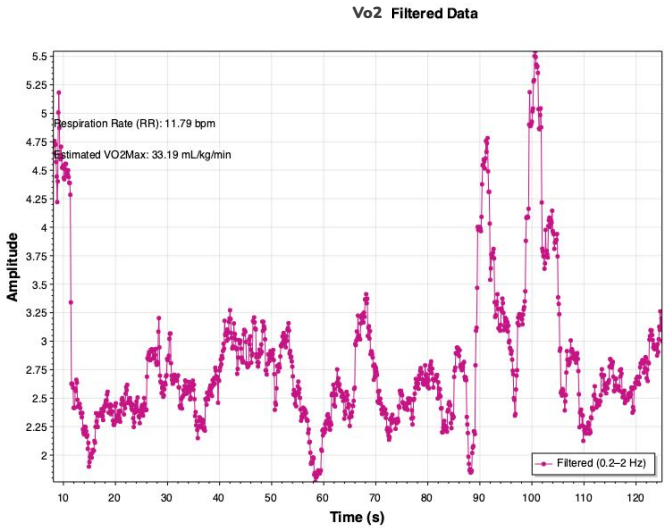
- This method first selects the data within the startTime and endTime. The signal is smoothed based on the effort level, and a peak detection algorithm figures out the peaks in the amplitude data (aka individual breaths). There was a minimum expected time between breaths factor added to dismiss peaks that are most likely not breaths. The RR is calculated by dividing the number of peaks by the total duration of the analysis window and converting the count to a rate of breaths per minute.

Data Filtering Methods

Calculating VO2Max: After getting the RR calculation, we used the Jackson et al. non-exercise model, which uses the age, gender, and BMI of the participant to figure out a VO2 value. The Physical Activity Level (PAL) multiplier was applied to that VO2 value based on the frequency and level of exercise the participant does. We added RR as a factor for the VO2Max estimation, and a lower boundary was added to stop the scaling factor from being negative.

Jackson et al. non-exercise model

```
//Non-exercise VO2Max model (Jackson et al.)
double vo2 = 50.513
             + (1.589 * genderFactor)
             - (0.289 * age)
             - (0.552 * bmi);
return vo2;
```



Result

Difference between ground truth and our program's estimated RR:

- An average difference of 0.73 has been seen so far in our data, which is very promising.

Ground Truth vs. Calculated RR

Camille:	Joseph:	Alyssa:	Thandar:
Light: 11 bpm vs. 12.40 bpm Moderate: 15 bpm vs. 16.36 bpm Heavy: 19 bpm vs. 19.83 bpm	Light: 13 bpm vs. 14.50 bpm Moderate: 17 bpm vs. 17.98 bpm Heavy: 21 bpm vs. 22.18 bpm	Light: 12 bpm vs. 12 bpm Moderate: 16 bpm vs. 15.93 bpm Heavy: 22 bpm vs. 21.36 bpm	Light: 10 bpm vs. 10.34 bpm Moderate: 14 bpm vs. 15.12 bpm Heavy: 19 bpm vs. 19.67 bpm
Average difference: +1.20 bpm	Average difference: +1.22 bpm	Average difference: -0.24 bpm	Average difference: 0.24 bpm

Reflection

What worked well

- Audio amplitude was a strong RR indicator
- Bluetooth mic provided consistent signal
- Short 2-minute were easier and cleaner to analyze

Challenges

- Noise (Wind + motion) affected signal
- Smartwatch VO2 max varied by brand (some don’t even record)

Reflection Cont:

What We Would Improve

- Larger Participant pool
- Compare against lab-grade RR if possible

Conclusion

- Microphone cannot fully replace VO2 max testing
  - Additional details are required, such as heart rate, age, and weight
- But smartphone audio is promising for RR tracking and may offer simple fitness insights
- Great potential for low-cost health monitoring